جامعة العلوم التطبيقية الخاصة
كلية تكنولوجيا المعلومات
الواجب الأول 2023-2

APPLIED SCIENCE UNIVERSITY
(ASU)
FACULTY OF INFORMATION TECHNOLOGY
(FIT)
FIRST HOMEWORK 2023-2

Subject: *Object oriented Programming 1301108*
Instructor: *Lawar Al Tawil*.....

Due: May. 10, 2024

Student Name: .....
Student ID: .....
☐ CS ☐ SE ☐ CCC ☐ DSAI ☐ Other

PLEASE NOTE THAT THIS HOMEWORK IS 1 PAGES AND WITH A TOTAL OF 10 POINTS

A) Animal Interface

Design a Java interface named Animal that contains the following methods:

1. int **getCurrentEnergy**(): This method returns the current energy level of the animal.
2. int **getHungerLevel**(): This method returns the current hunger level of the animal.

B) Bird Class

Write a class called Bird that implements the Animal interface. This class should include the following:

1. Two public integer variables named **currentEnergy** and **hungerLevel**.
2. Implementation of the **getCurrentEnergy**() method to return the current energy level.
3. Implementation of the **getHungerLevel**() method to return the current hunger level.
4. A method named **fly**() that:
   - Sets the **currentEnergy** to 50 units.
   - Decreases the **hungerLevel** by 2 units.

C) Demo Class

In a test class named Demo, perform the following:

1. Create an instance of the Bird class with initial values for **currentEnergy** and **hungerLevel**.
2. Call the **fly()** method on the Bird instance.
3. Print the current energy level and hunger level of the Bird instance after calling the **fly**() method.

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A

```
Public interface LightSensor {
    Public Void turnOn (boolean status);
    Public Void adjustSensitivity (int level);
}
```

B

```
Public class AmbientLightSensor implements LightSensor {
    Public int sensitivity;
    Public Void turnOn (boolean status){
        if (status == true)
            System.out.Println ("Light Sensor is now on");
        else
            System.out.Println ("Light Sensor is now off");
    }
    Public void adjustSensitivity( int level){
        this. Sensitivity = level;
        System. out. Println ("Sensitivity set to X");
    }
}
```

C

```
Public class TestLightSensor {
    Public static void main (String [] args){
        AmbientLightSensor X = new AmbientLightSensor( );
        X. turnOn ( true);
        X. adjustSensitivity (7);
        System. out. println ("Sensitivity Level: " + X. sensitivity);
    }
}
```

Quiz 2 2023-2

Subject: *Object Oriented Programming 1301108*

Instructor: *Dr. Mahmoud S. Jawarneh*.....

Due: 21 May, 2024

Student Name: .....

Student ID: .....

☐ CS ☐ SE ☐ CCC ☐ DSAI ☐

PLEASE NOTE THAT THIS QUIZ IS 1 PAGE AND WITH A TOTAL OF 10 POINTS.

QUESTION 1: [10 PO

A. Design a java interface named **Playable** that contains the following:

- A method called ***playWith***(*int players*). This method takes an integer parameter tha represents the number of players in the game.
- A method named ***setResult***(*String result*). This method takes a String parameter t represents the result of the game (*Win, Loss, or Tie*).

B. Write a class called **Football** that implements the **Playable** interface, this class includes following:

- A public String variable is named ***result***.
- ***playWith***(*int players*): This method prints " Football Team is completed." if the players is 11, or " Football Team is not completed" if the number of players is
- ***setResult***(*String result*): This method sets the property **result** to the provided

C. In a test class named **TestPlay** do the following:

- Create an object for the **Football** class.
- Call ***playWith method*** through declared object sending a number of players
- Call ***setResult*** method through send "*Win*" as an argument.
- Print the ***result*** value.

Write your code here:

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كلية تكنولوجيا المعلومات
امتحان قصير 2023-2

Subject: *Object Oriented Programming 1301108*
Instructor: *Dr. Mansour Al-Helalat*.....
.....
.....

Due: 21 May. 2024

Student Name [redacted]
Student ID [redacted]
☐ CS ☐ SE ☐ CCC ☑ DSAI ☐ Other

PLEASE NOTE THAT THIS QUIZ IS 1 PAGE AND WITH A TOTAL OF 10 POINTS.

QUESTION 1: [10 POINTS]

A. Design a java interface named **Sport** that contains the following:

- A method called **startStopGame**(boolean status), which, when implemented, allows objects to start or stop a game. This method receives a boolean parameter indicating whether the game is started (true) or stopped (false).
- A method named **setTime** (int remainingTime), which, when implemented, adjusts the time of the game. This method accepts an integer parameter that specifies the remaining time of the game.

B. Write a class called **Football** that implements the **Sport** interface, this class includes the following:

- A public integer variable named **time**.
- **startStopGame**(boolean status): This method should control whether the football match is started or stopped. It prints "Match is now started." if true, or "Match is now stopped." if false.
- **setTime**(int remainingTime): it sets the **time** to the remaining time.

C. In a test class named TestGame do the following:

- Create an instance of the **Football** class.
- Test **startStopGame** Functionality: Use the startStopGame method to start the game (startStopGame(**true**)).
- Test **setTime**: Adjust the time to 75.
- Print the game **time**.

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كلية تكنولوجيا المعلومات
امتحان قصير 2023-2

Subject: *Object Oriented Programming 1301108*
Instructor: *Radwan Batyha*.....
.....
.....

Due: 21 May. 2024

Student Name: .....
Student ID: .....
□ CS □ SE □ CCC □ DSAI □ Other

PLEASE NOTE THAT THIS QUIZ IS 1 PAGE AND WITH A TOTAL OF 10 POINTS.

QUESTION 1: [10 POINTS]

A. Design a java interface named **UnivGrader** that contains the following:

- A method called **isPassed**(), which returns **boolean**, when implemented, return true if the mark is passed mark or false if the mark is failed mark depending on the class that implements it (**UnderGradMark** or **PostGradMark**).

B. Write a class called **UnderGradMark** that implements the **UnivGrader** interface, this class includes the following:

- A public integer variable named **mark**.
- A parameterized constructor that receives a mark.
- **isPassed**(): This method should determine if the mark is passed or fail (>=50 ➔ Pass, <50 ➔ Fail).

C. In a test class named TestMark do the following:

- Create one instance of the **UnderGradMark** classes.
- Test if the mark is passed or failed.
- The sample runs below to show the inputs and outputs.

SAMPLE RUN:

```
Enter Mark for Under Grad : 70
70 is Pass Mark
```

```
Enter Mark for Under Grad : 45
45 is Fail Mark
```

// WRITE YOUR CODE HERE

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Quiz ٢ 2023-2

جامعة العلوم التطبيقية الخاصة
كلية تكنولوجيا المعلومات
امتحان قصير ٢٠٢٣-2

Subject: *Object Oriented Programming 1301108*
Instructor: *I.Barsa Qadoumi*.....
.....
.....

Due: 22 May. 2024

10

Student Name: .....
Student ID: .....
□ CS □ SE □ CCC □ DSAI □ Other

PLEASE NOTE THAT THIS QUIZ IS 1 PAGE AND WITH A TOTAL OF 10 POINTS.

QUESTION 1:

[10 POINTS]

A. Design a Java interface named LightSensor that contains the following:

- A method called turnOn(boolean status), which, when implemented, turns the light sensor on or off. This method receives a boolean parameter indicating whether the sensor is turned on (true) or off (false).
- A method named adjustSensitivity(int level), which, when implemented, adjusts the sensitivity level of the light sensor. This method accepts an integer parameter that specifies the desired sensitivity level.

B. Write a class called AmbientLightSensor that implements the LightSensor interface. This class includes the following:

- A public integer variable named sensitivity.
- turnOn(boolean status): This method should control whether the sensor is turned on. It prints "Light Sensor is now on." if true, or "Light Sensor is now off." if false.
- adjustSensitivity(int level): This method should set the sensitivity to the specified level and print "Sensitivity set to X," where X is the specified level.

C. In a test class named TestLightSensor, do the following:

- Create an instance of the AmbientLightSensor class.
- Test turnOn functionality: Use the turnOn method to turn on the sensor (turnOn(true)).
- Test adjustSensitivity: Adjust the sensitivity to a level of 7.
- Print the sensitivity level after each operation.

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Quiz 2 2023-2

كلية تكنولوجيا المعلومات
امتحان قصير 2-2023

10

Subject: *Object Oriented Programming 1301108*
Instructor: *I.Hanan Hardan*

Due: 21 May. 2024

Student Nam[redacted]
Student ID: [redacted]
□ C[redacted]

PLEASE NOTE THAT THIS QUIZ IS 1 PAGE AND WITH A TOTAL OF 10 POINTS.

[10 POINTS

QUESTION 1:

A. Design a Java interface named **Player** that contains the following:

1. A method called **setActive(boolean status)**, which, when implemented, allows objects to activate or deactivate the player. This method receives a boolean parameter indicating whether the player is active (true) or inactive (false).
2. A method named **setScore(int points)**, which, when implemented, adjusts the score of the player. This method accepts an integer parameter that specifies the number of points.

B. Write a class called **BasketballPlayer** that implements the **Player** interface. This class includes the following:

- A public integer variable named **score**.
- **setActive(boolean status)**: This method should control whether the player is active. It prints "Player is now active." if true, or "Player is now inactive." if false.
- **setScore(int points)**: This method sets the score to the specified value.

C. In a test class named **TestPlayer**, do the following:

1. Create an instance of the **BasketballPlayer** class.
2. Test the active functionality: Use the **setActive** method to activate the player (setActive(true)).
3. Test the score: Adjust the score to 15 using the **setScore** method.
4. Print the score.

```
C. public class TestPlayer{
    public static void main (String [ ] args){
        BasketballPlayer b1 = new BasketballPlayer ();
        b1.setActive (true);
        b1.setScore (15);
        System.out.println (b1.score); }}
```

2

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A. public interface Player {

```
    public String setActive(boolean status);
    public void setScore(int points);
}
```

1.75

B. public class BasketballPlayer implements Player { 0.25

```
    public int score; 0.25
    @Override
    public String setActive(boolean status) { 0.75
        if(status==true) 0.5
            return "Player is now active"; 0.25
        else if (status==false) 0.25
            return "Player is now inactive"; 0.75
        else
            return null;
```

3.5

```
    @Override
    public void setScore(int points) {
        score=points;
```

البرمجة الكينونية (1) - 2

Question **1**
Not yet answered
Marked out of 10.00
Flag question

Time left 0:39:30

Quiz navigation

1

Finish attempt ...

A. Design a class named **Book** that contains the following:

- An instance variable **price** (double).
- An instance variable **format** (String) to represent the book's format (e.g., "Paperback", "Kindle").
- A Parameterized constructor to initialize the instance variables with the values provided.
- An instance method **getNetPrice()** that calculates and **returns** the Book's net price based on its format with the following criteria:

  *If the book format is "Paperback", then netPrice = price * 0.9, if the book format is "Kindle", then netPrice = price * 0.7, otherwise if the book does not fall into either category, return 0.*

B. In a test class named **BookTester**, prompt the user to input the book's details: **price** and **format**, instantiate an object of the **Book** class with the values entered by the user, then output the calculated **netPrice** on the screen.

Paste your answer below each header

*A. Paste the Class **Book** Code below this line:*

```
package
```

ENG

ThinkVision

Time left 0:38:24

A. Design a class named **Vehicle** that contains the following:

- An instance variable **fuelCapacity** (integer) to represent the total fuel a vehicle can hold.
- An instance variable **type** (String) to represent the vehicle's type (e.g., "Electric", "Gasoline").
- A parameterized constructor to initialize the instance variables with the values provided.
- An instance method **getRange()** that calculates and **returns** the Vehicle's travel range based on its type with the following criteria:

  *If the vehicle type is "Electric", then range = fuelCapacity * 10, if the vehicle type is "Gasoline", then range = fuelCapacity * 15, otherwise if the vehicle does not fall into either category, return 0.*

B. In a test class named **TestDrive**, prompt the user to input the vehicle's details: **type** and **fuelCapacity**, instantiate an object of the **Vehicle** class with the values entered by the user, then output the calculated travel range on the screen.

Paste your answer below each header

*A. Paste the Class **Vehicle** Code below this line:*

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امتحان قصير 2023-2

Subject: *Object Oriented Programming 1301108*
Instructor: *I.Hadeel Ahmad*.....

Due: 21 May, 2024

Student Name: .....
Student ID: .....
☐ CS ☐ SE ☐ CCC ☐ DSAI ☐ Other

PLEASE NOTE THAT THIS QUIZ IS 1 PAGE AND WITH A TOTAL OF 10 POINTS.

QUESTION 1: [10 POINTS]

A. Design a java interface named **SoundEmitter** that contains the following:

- A method called **mute**(boolean status), which, when implemented, allows objects to mute or unmute the sound. This method receives a boolean parameter indicating whether the sound is muted (true) or unmuted (false).
- A method named **setVolume** (int level), which, when implemented, adjusts the volume level of the sound the object emits. This method accepts an integer parameter that specifies the desired volume level.

B. Write a class called **Bell** that implements the **SoundEmitter** interface, this class includes the following:

- A public integer variable named **volume**.
- **mute**(boolean status): This method should control whether the bell can make a sound. It prints "Bell is now muted." if true, or "Bell is now unmuted." if false.
- **setVolume**(int level): it sets the **volume** to the specified **level**.

C. In a test class named TestSound do the following:

- Create an instance of the **Bell** class.
- Test **Mute** Functionality: Use the mute method to mute the bell (mute(**true**)).
- Test **Volume**: Adjust the volume to a level of **8**.
- Print the **volume** level.

```
public class TestSound {
    public static void main (String[] args) {

        Bell x = new Bell();
        x.mute(true);
        x.setVolume(8);
        s.out(x.volume);
    }
}
```

A)
```
public interface SoundEmitter {

    public void mute( boolean m);
    public void setVolume (int v);
}
```

B)
```
public class Bell implements SoundEmitter {
    public int volume;

    @Override
    public void mute (Boolean m) {
        if(m== true) {
            s.out(" Bell is now muted");
        }
        else
            s.out(" Bell is now unmuted");
    }

    ~~public~~
    @Override
    public void setVolume (int v) {
        this.volume =v;
    }
```